



Introduction to *AWS*

By: Ahmed Samir



Meet Your Instructor

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Meet Your Instructor

Certificates:

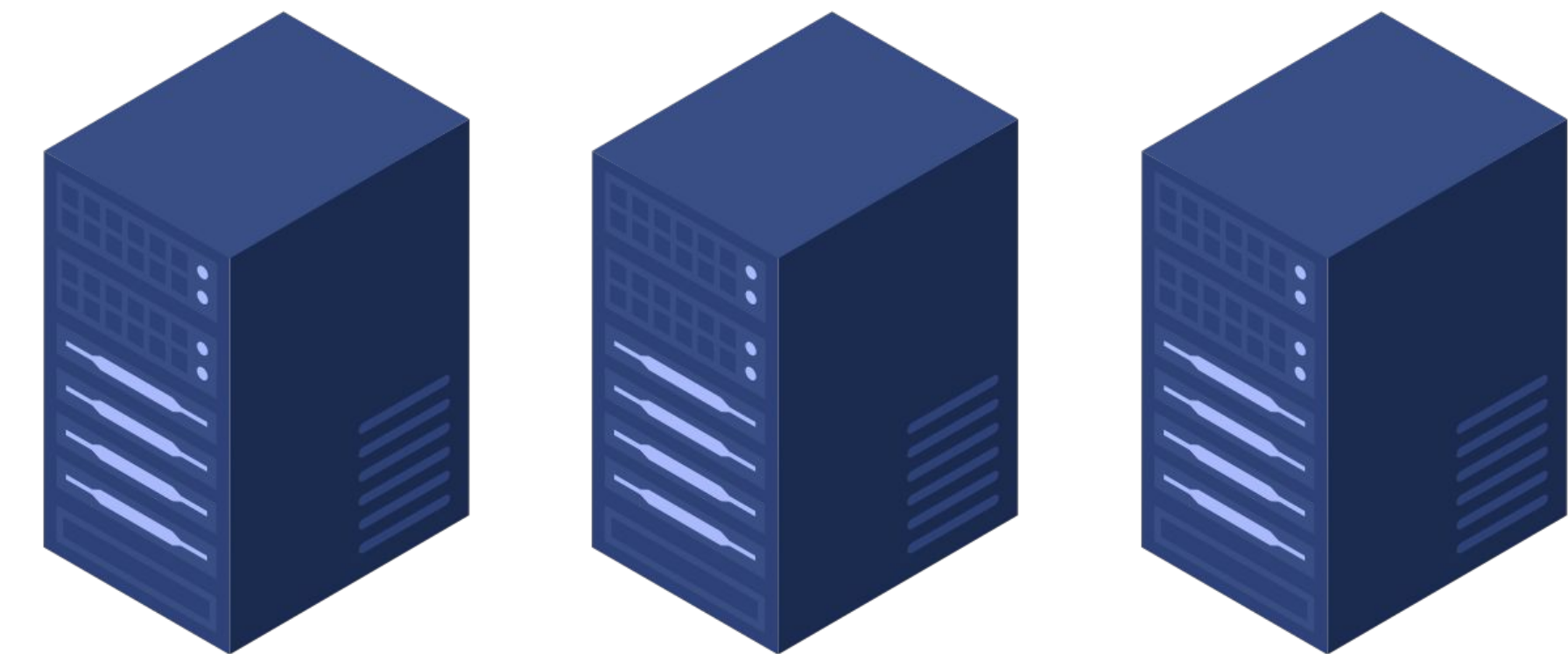
- AWS Certified Security Specialty
- AWS Data Analytics Specialty
- AWS Solutions Architect Associate
- Certified Kubernetes Administrator (CKA)
- Microsoft Azure Administrator Associate
- Trend Micro Advanced Threat Defense: Ethical hacking and Incident response
- Trend Micro Deep Security
- Trend Micro Apex One

Cloud Computing

What is Cloud Computing?

Before Cloud Computing (On-premises data center)

- Servers
 - Web Servers, DB Servers, ...
- CPU
- RAM
- Storage
- Network + Internet connection
- Security
 - Firewalls, IDPS, ...



What is Cloud Computing?



Cloud Global Infrastructure

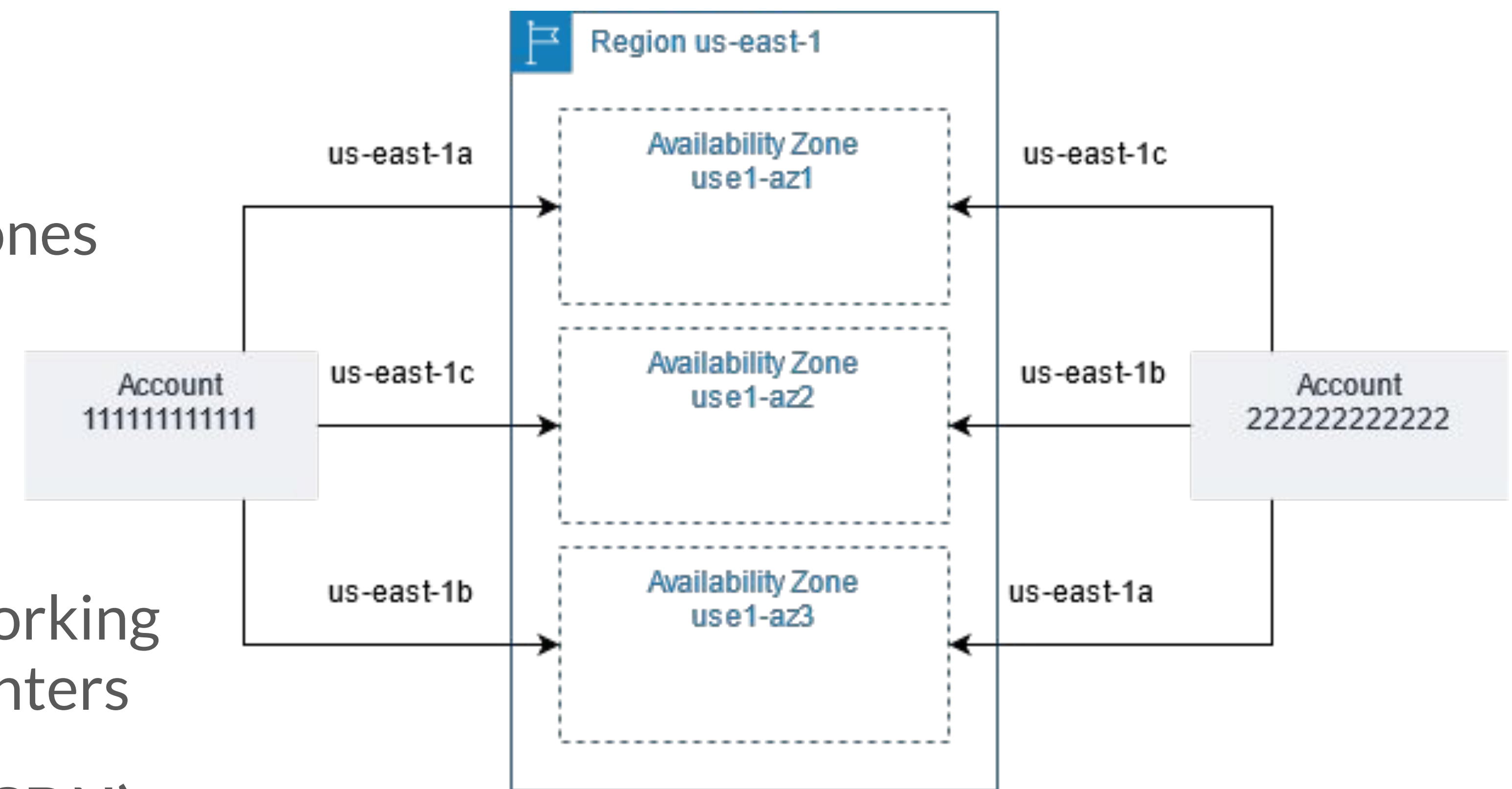
- Regions

- Isolated geographic area
- Each Region has at least 3 Availability Zones

- Availability Zones

- Independent locations
- Connected through low-latency, high-bandwidth, highly-redundant networking
- Consists of one or more discrete data centers

- Edge Locations (Content Delivery Network CDN)



https://aws.amazon.com/about-aws/global-infrastructure/regions_az/



Cloud Global Infrastructure

AWS announces new edge location in Egypt

Posted on: May 22, 2024

Amazon Web Services (AWS) announces expansion in Egypt by launching a new Amazon CloudFront edge location in Cairo, Egypt. Customers in Egypt can expect up to 30% improvement in latency, on average, for data delivered through the new edge location. The new AWS edge location brings the full suite of benefits provided by [Amazon CloudFront](#), a secure, highly distributed, and scalable content delivery network (CDN) that delivers static and dynamic content, APIs, and live and on-demand video with low latency and high performance.

<https://aws.amazon.com/about-aws/whats-new/2024/05/new-edge-location-egypt/>



Why we needed Cloud Computing?

Characteristics:

- Scalability
- Reliability
- Elasticity
- Agility
- Security
- Cost (Pay-as-you-go)



Cloud Services Models

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

 You manage

 Service provider manages

redhat.com



Top Cloud Providers

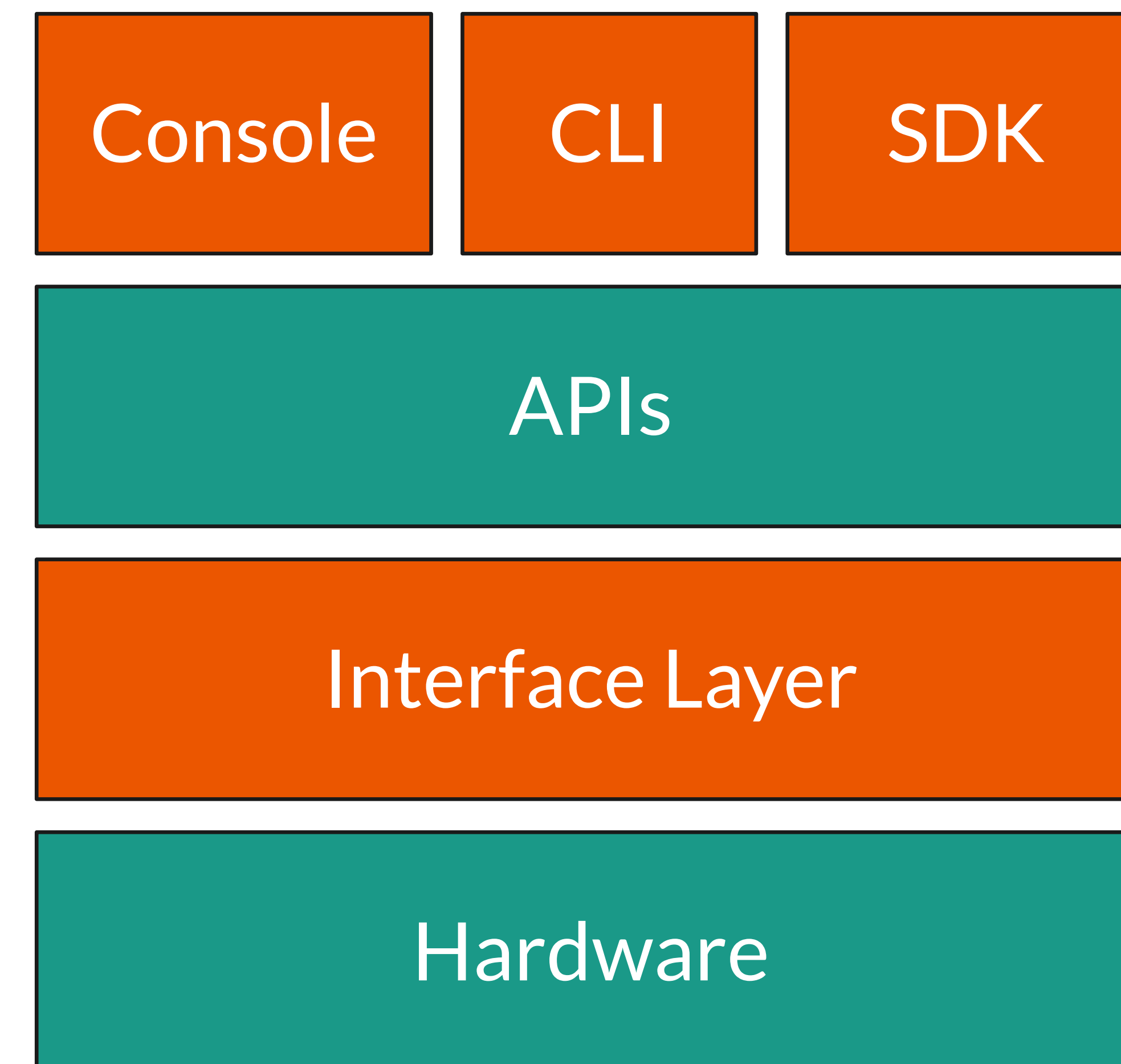
- Amazon Web Services
- Microsoft Azure
- Google Cloud
- IBM Cloud
- Oracle Cloud
- Alibaba Cloud



AWS Resources Management Tools

Everything is an API

- AWS Management Console
- AWS CLI
- AWS SDK (Python, C#, Java, ...)



AWS Services

 Analytics	 Application integration	 Blockchain	 Business applications	 Cloud Financial Management	 Compute
 Customer enablement	 Containers	 Databases	 Developer tools	 End user computing	 Front-end web and mobile
 Game tech	 Internet of Things (IoT)	 Machine Learning (ML) and Artificial Intelligence (AI)	 Management and governance	 Media	 Migration and transfer
 Networking and content delivery	 Quantum technologies	 Satellite	 Security, identity, and compliance	 Storage	



AWS Services



Virtual Private
Cloud (**VPC**)



Elastic Cloud
Compute (**EC2**)



Elastic Block
Storage (**EBS**)



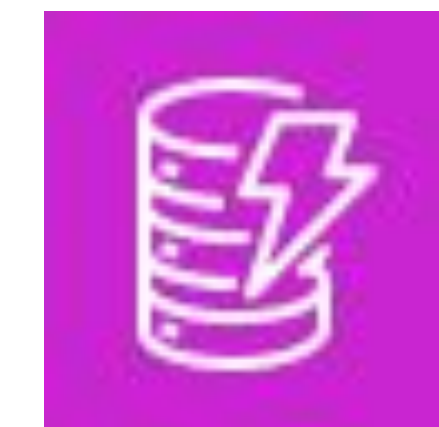
Simple Storage
Service (**S3**)



Identity and Access
Management (**IAM**)



Relational Database
Service (**RDS**)



DynamoDB



SageMaker

AWS Networking

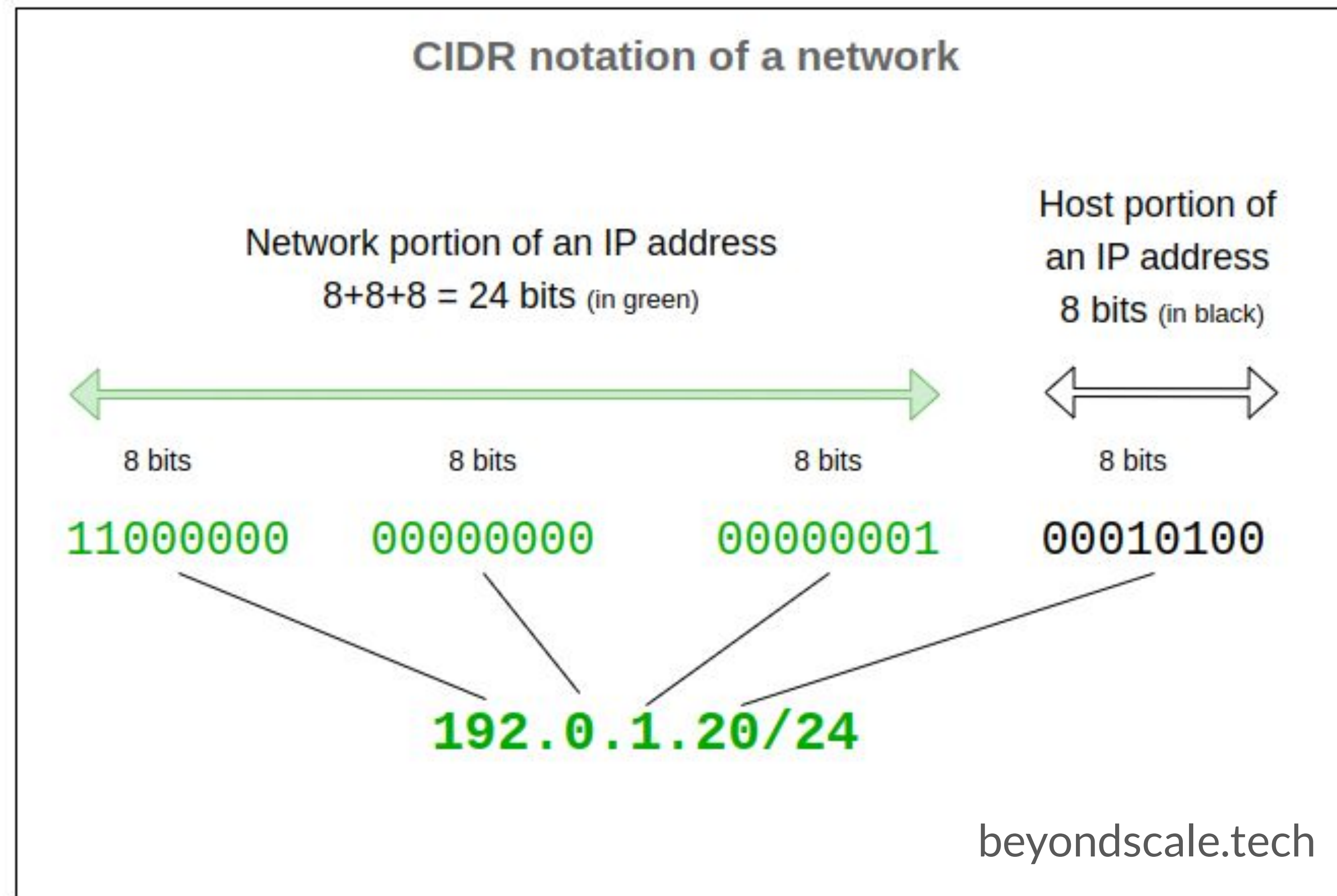
AWS Networking

- VPC
- Subnets
- Route Tables
- Gateway
 - Internet GW
 - NAT GW



Virtual Private Cloud (VPC)

CIDR





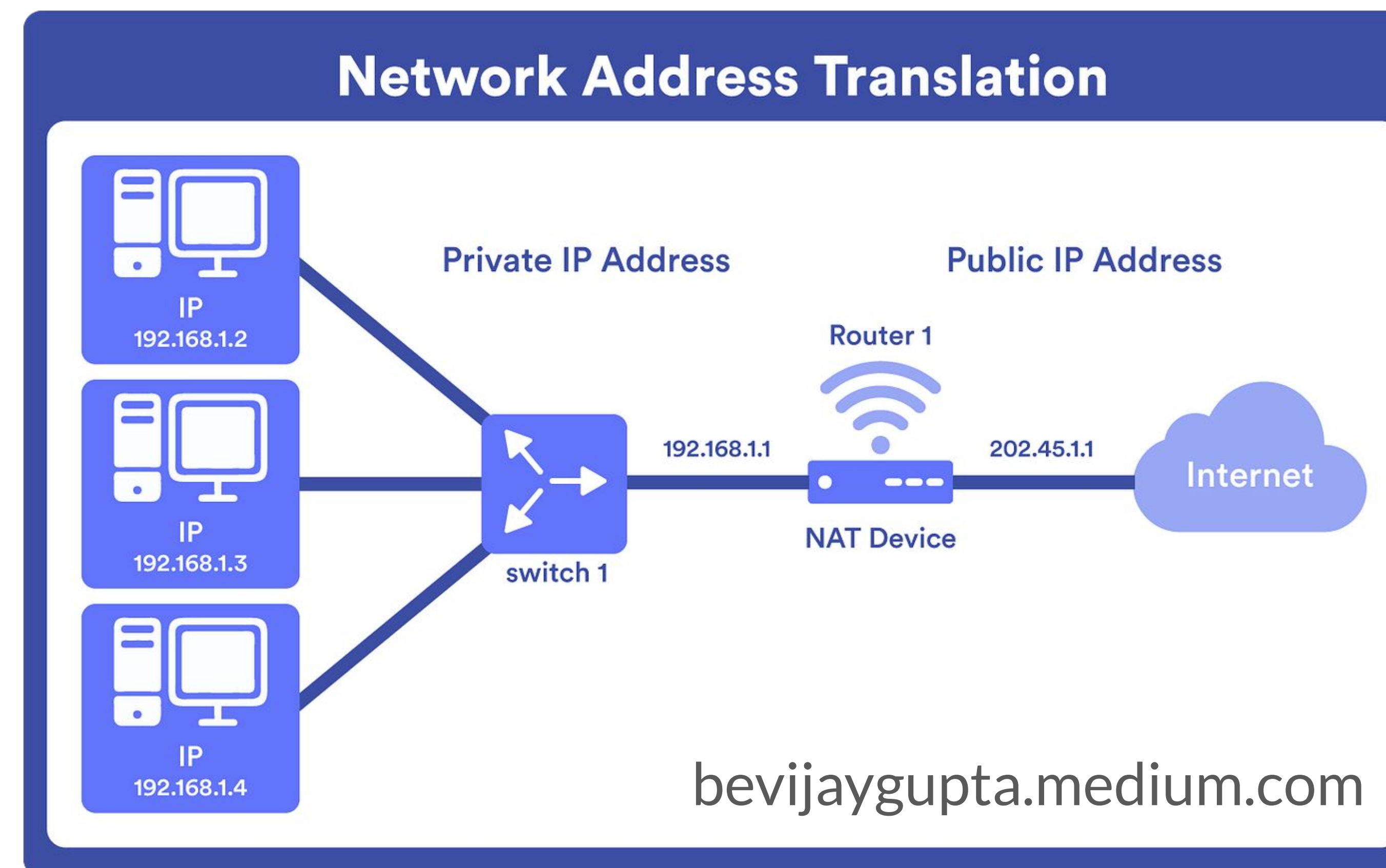
IP Addressing

- Public IP: Exposed to the internet
- Private IP: Not exposed to the internet (in private network)

Private IP Address Ranges
10.0.0.0 - 10.255.255.255
172.16.0.0 - 172.31.255.255
192.168.0.0 - 192.168.255.255

[linkedin.com/in/charlescrampton](https://www.linkedin.com/in/charlescrampton)

Network Address Translation (NAT)



AWS Network Design Diagram

Route Table

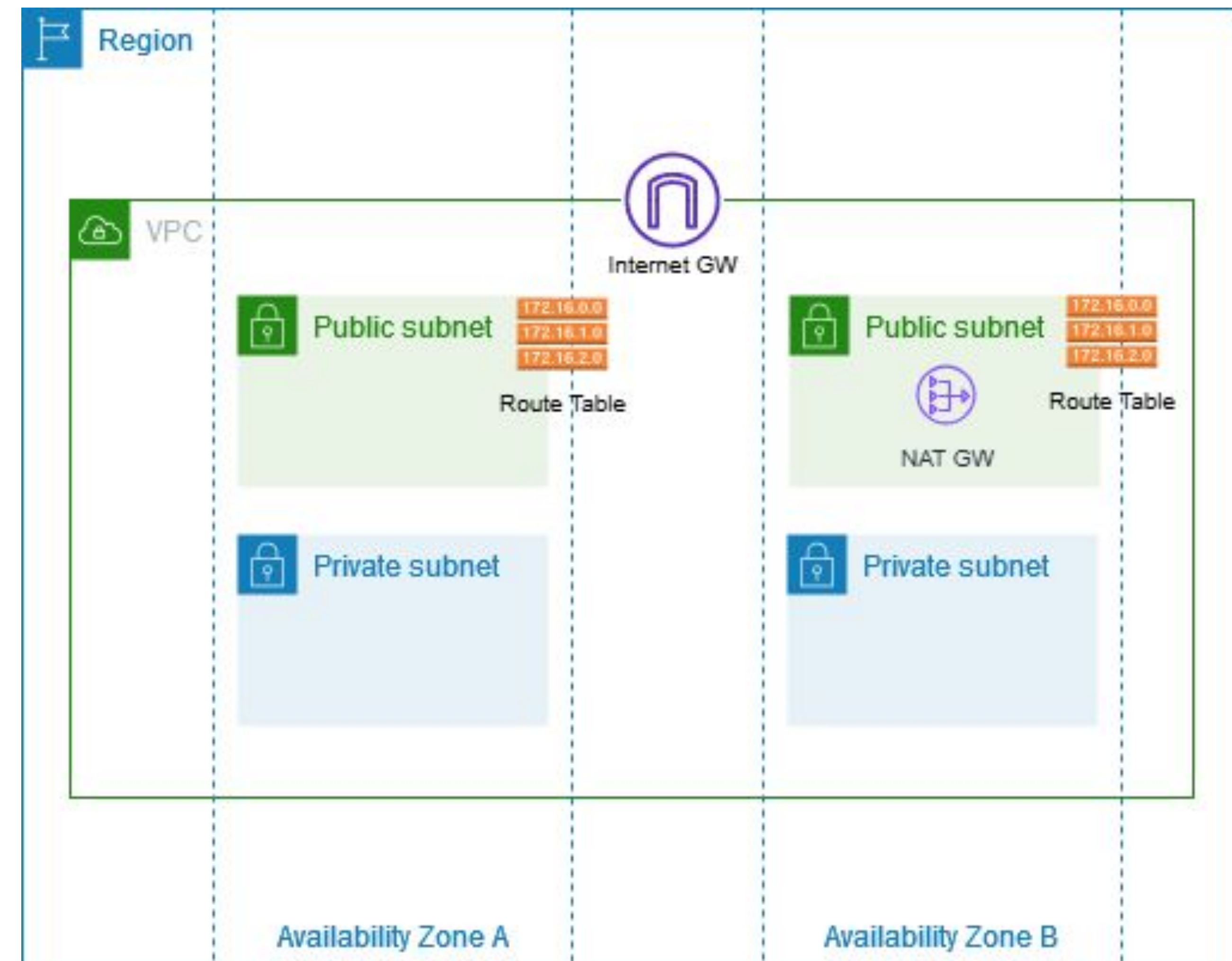
- Set of rules for network traffic routing

Internet Gateway

- Internet access for the VPC
- Inbound / Outbound

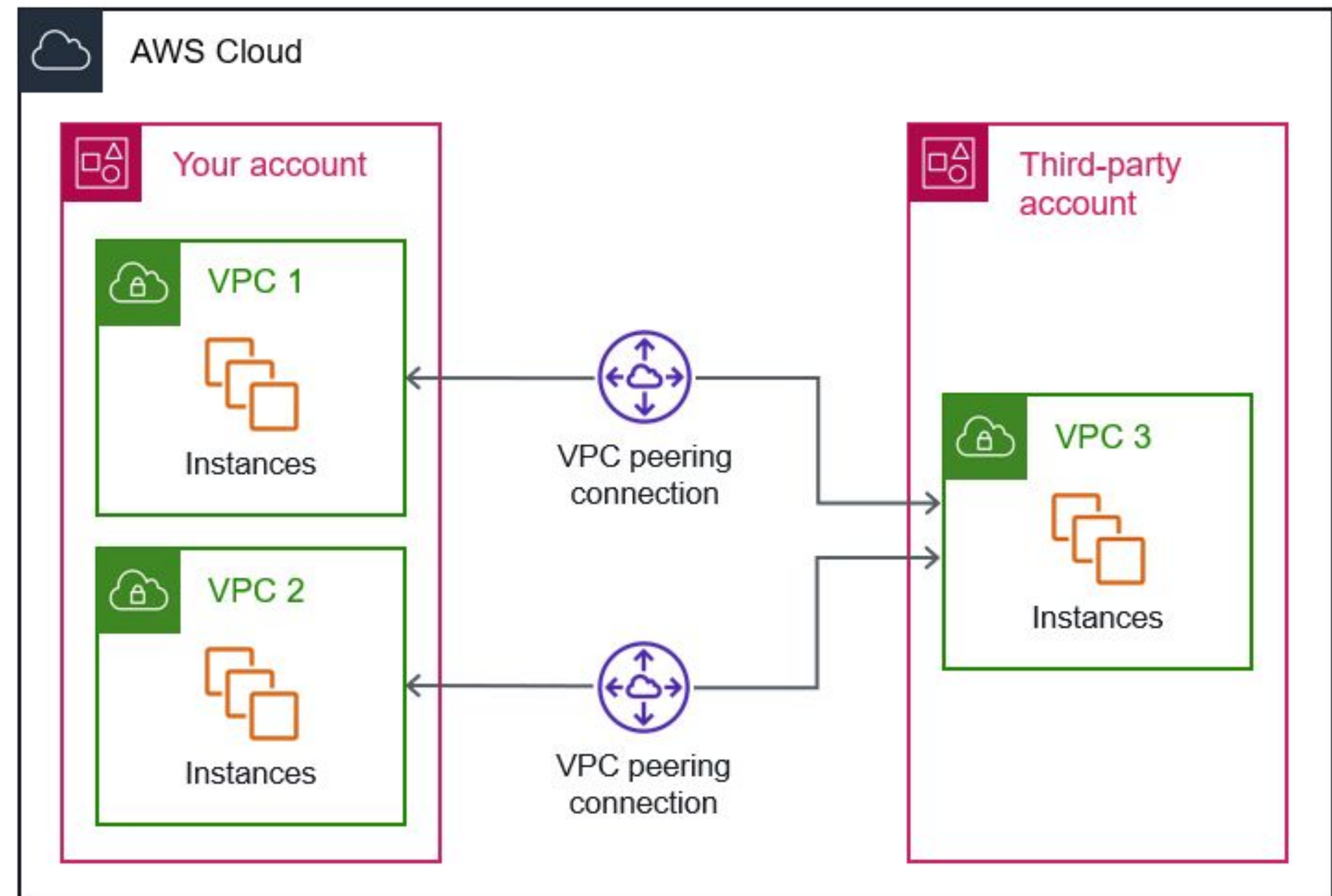
NAT Gateway

- Internet access for private resources
- Uses IGW to get to the Internet
- Outbound only



VPC Peering

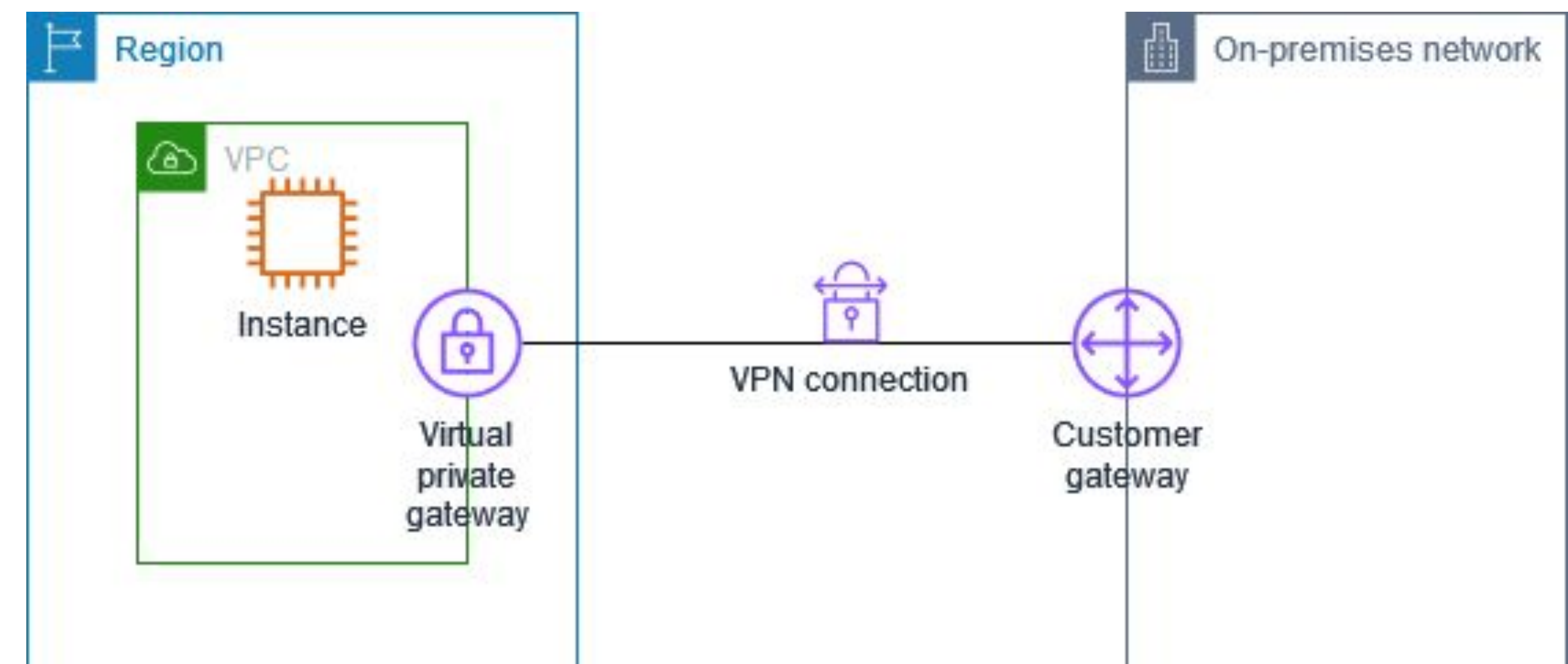
- Connect 2 VPCs privately
- VPCs can be in different accounts
- VPCs can be in different Regions



Site-to-Site VPN

A secure connection between your on-premises network and your VPC

- **Virtual Private Gateway** attached to a VPC
- **Customer gateway device** is a physical device or software application on the on-premises side
- **Customer gateway** resource on AWS to connect to the device



AWS Elastic Compute Cloud (EC2)



What is EC2?

It's a VM!

- CPU
- RAM
- Storage
- OS





EC2 Pricing

Payment Options:

- **On-Demand:** Pay by the second
- **Spot Instances:** Request unused EC2 instances, reduces your costs significantly
- **Savings Plan:** Reduce your costs by making a commitment to a consistent amount of usage, in USD per hour, for a term of 1 or 3 years
- **Reserved Instances:** Reduce your costs by making a commitment to a consistent instance configuration, including instance type and Region, for a term of 1 or 3 years.
- **Dedicated Hosts:** Pay for a physical host that is fully dedicated to running your instances



EC2 Configuration

- Instance Types (CPU / RAM / Network Bandwidth)
- AMI (OS + Software apps installed on the machine)
- Network (VPC / Subnets / ...)
- Storage -> next slide
- Security Group (Inbound traffic / Outbound traffic) -> next slide
- Userdata (Startup script to install software / download files / ...)
- SSH Key Pair for SSH connection



EC2 Storage Types

EBS

- Persistent block storage that can be detached and attached to another instance
- You can create snapshots from EBS volumes as backups

Instance Store

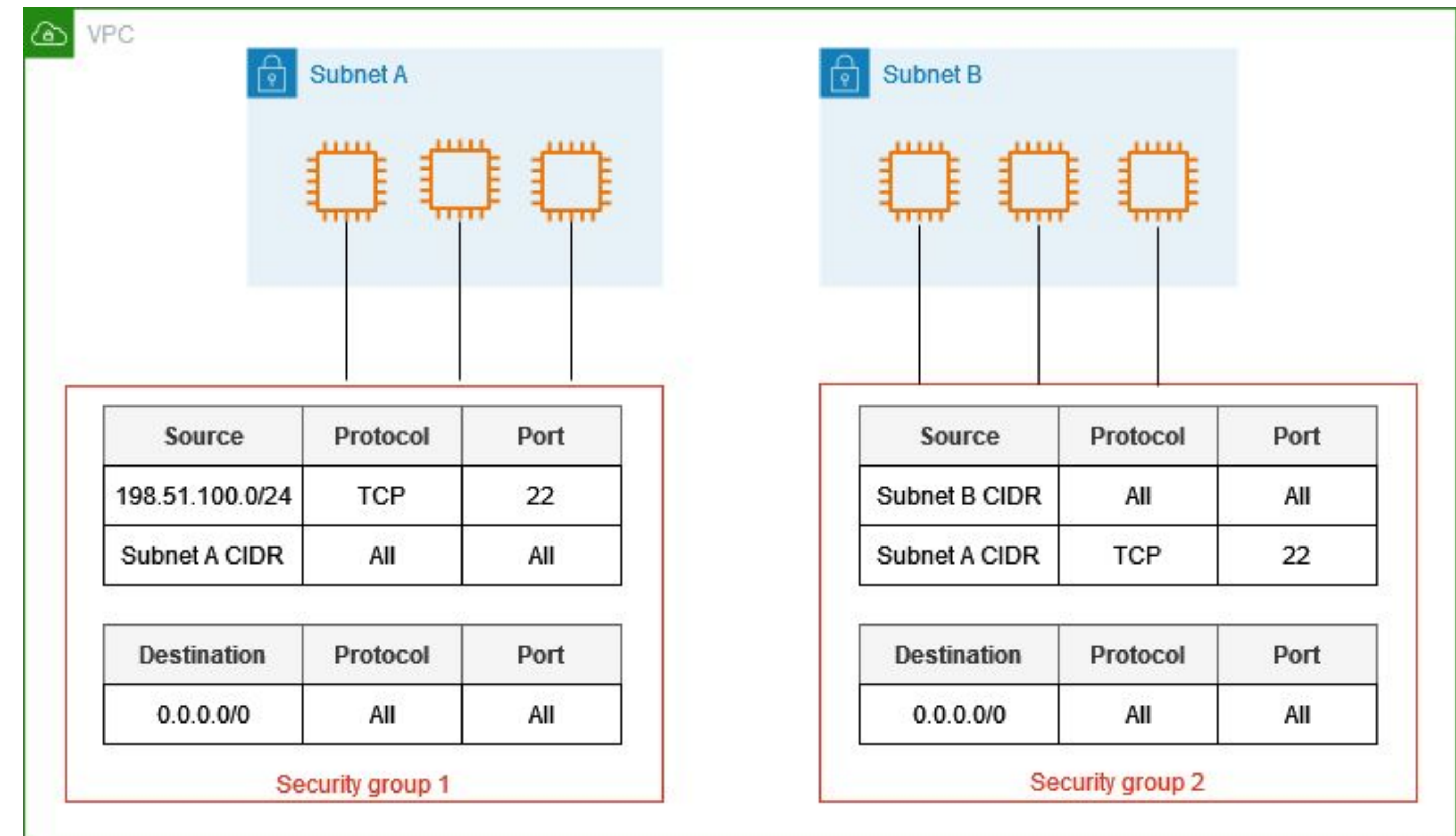
- Temporary block storage, persists only during the life of the associated instance

EFS (Linux) / FSx (Windows)

- Scalable file storage. You can configure your instances to mount the file system to be used as a shared data source for workloads running on multiple instances.

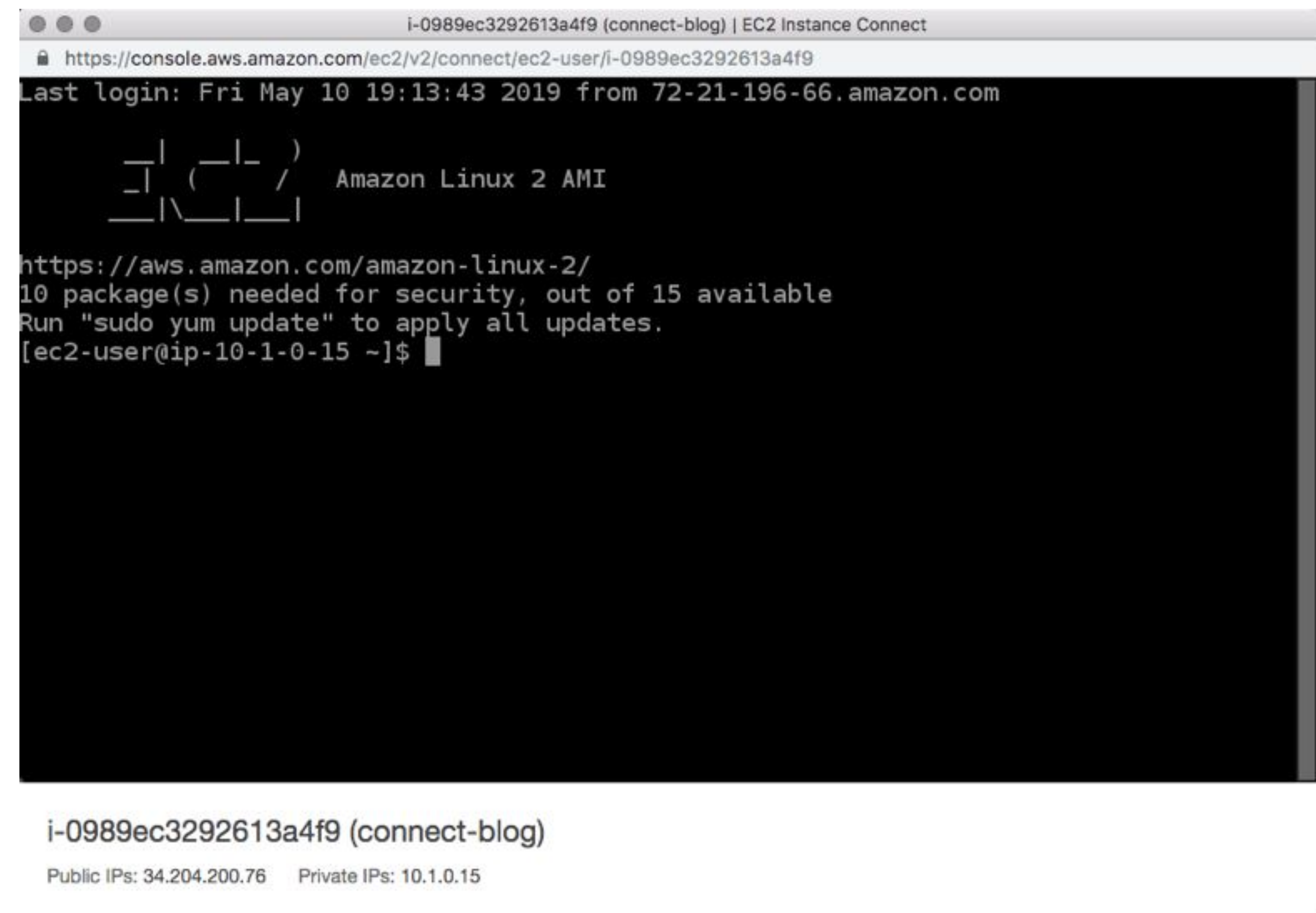
EC2 Security Groups

- EC2 Firewall to control inbound / outbound traffic
- Allow-only rules (not allowed is blocked)



Connect to EC2

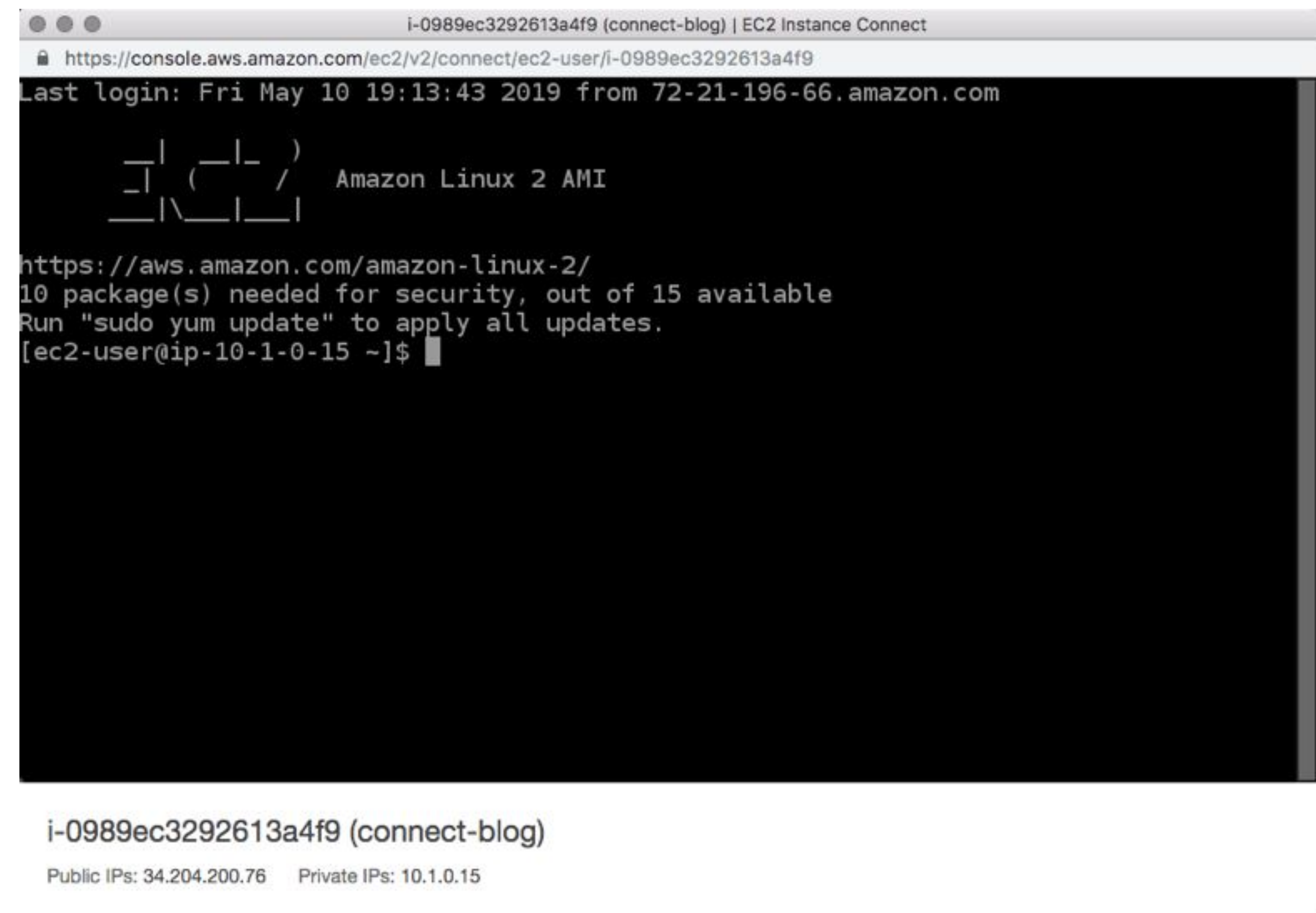
- Connect to public EC2 instance
 - EC2 connect
 - SSH
- Connect to private EC2 instance?



The screenshot shows a terminal window titled "i-0989ec3292613a4f9 (connect-blog) | EC2 Instance Connect". The browser address bar shows "https://console.aws.amazon.com/ec2/v2/connect/ec2-user/i-0989ec3292613a4f9". The terminal output includes the last login time "Last login: Fri May 10 19:13:43 2019 from 72-21-196-66.amazon.com", the Amazon Linux 2 AMI logo, and a message: "https://aws.amazon.com/amazon-linux-2/ 10 package(s) needed for security, out of 15 available Run 'sudo yum update' to apply all updates." The prompt is "[ec2-user@ip-10-1-0-15 ~]\$". At the bottom, the instance ID "i-0989ec3292613a4f9 (connect-blog)" and IP addresses "Public IPs: 34.204.200.76 Private IPs: 10.1.0.15" are displayed.

Connect to EC2

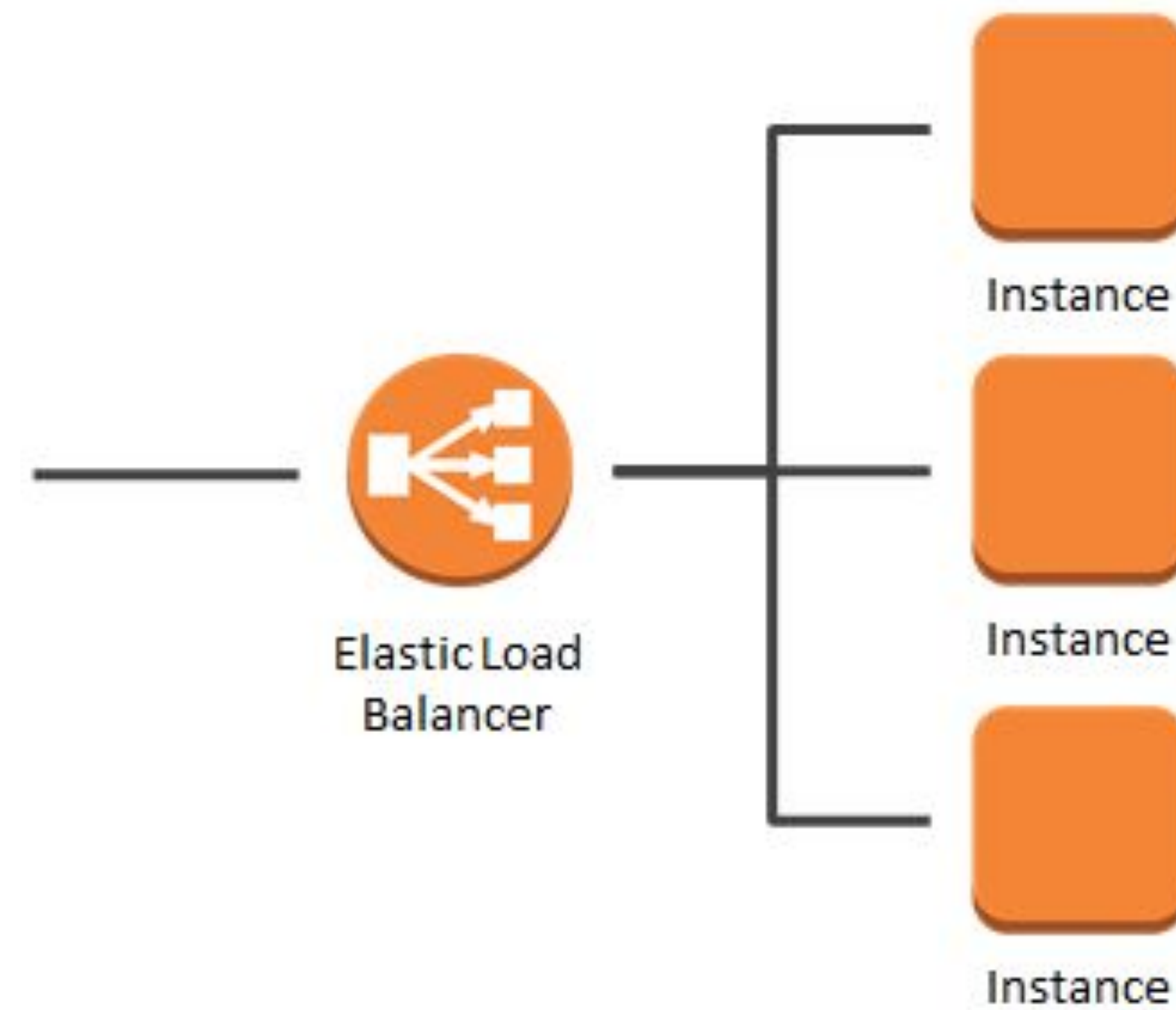
- Connect to public EC2 instance
 - EC2 connect
 - SSH
- Connect to private EC2 instance?
 - Jump Server / Bastion Host



The screenshot shows a terminal window titled "i-0989ec3292613a4f9 (connect-blog) | EC2 Instance Connect". The browser address bar shows the URL "https://console.aws.amazon.com/ec2/v2/connect/ec2-user/i-0989ec3292613a4f9". The terminal output includes the text "Last login: Fri May 10 19:13:43 2019 from 72-21-196-66.amazon.com", the Amazon Linux 2 AMI logo, and a message: "https://aws.amazon.com/amazon-linux-2/ 10 package(s) needed for security, out of 15 available Run 'sudo yum update' to apply all updates." The prompt is "[ec2-user@ip-10-1-0-15 ~]\$". At the bottom of the window, it displays "i-0989ec3292613a4f9 (connect-blog)" and "Public IPs: 34.204.200.76 Private IPs: 10.1.0.15".

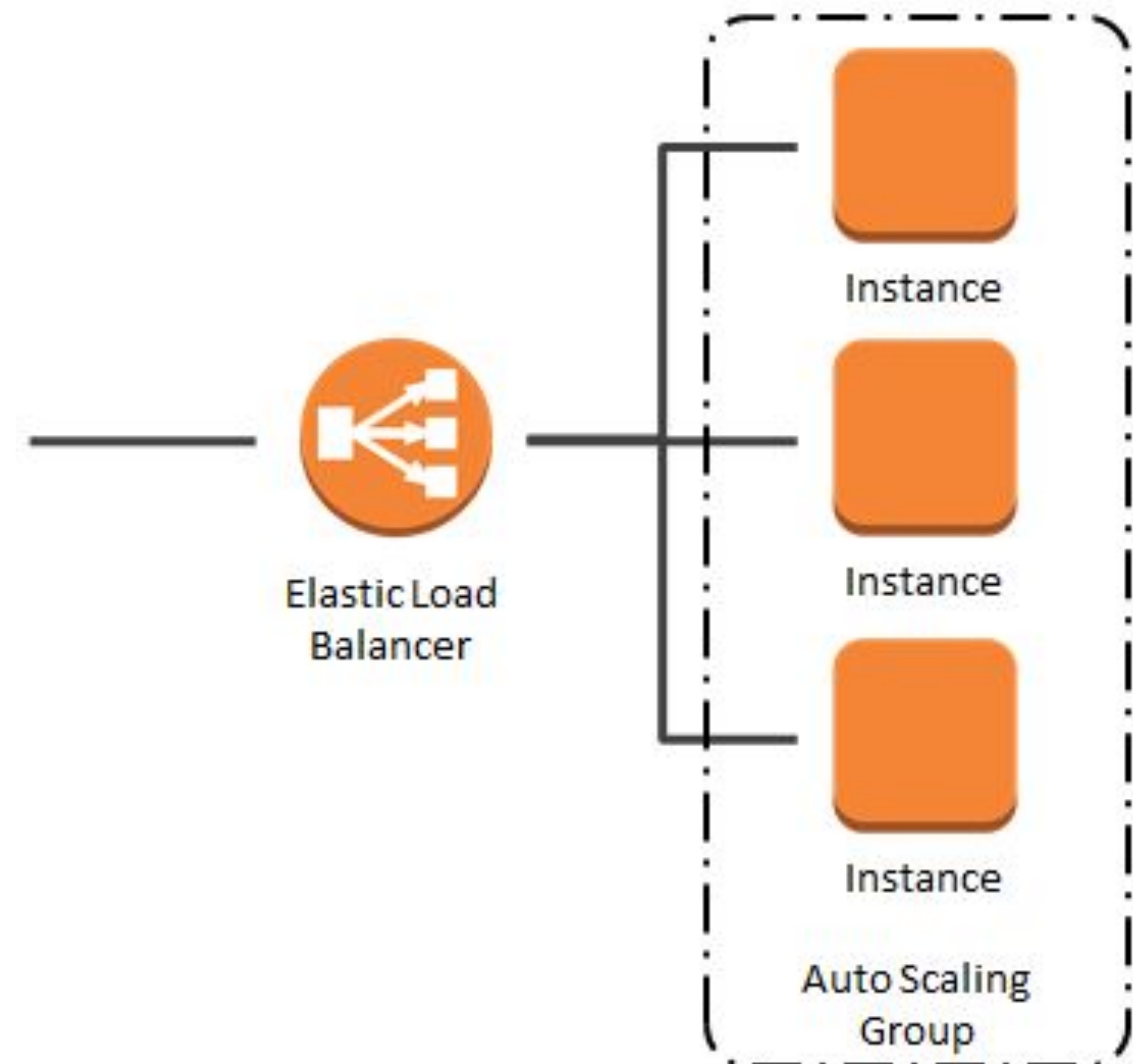
Elastic Load Balancer (ELB)

- Load balancing
- Routing
 - Path (app.com/users)
 - Hostname (dev.app.com)
- Health checks



EC2 Auto Scaling group (ASG)

- Scale in/out based on load
- Min / Max / Desired
- Different metrics for scaling in/out
 - Schedule (Friday @ 12PM)
 - Alarm (CPU usage > 75%)
- Launch Template: store instance launch parameters so that you don't have to specify them every time you launch an EC2 instance



—

Lambda

Lambda

- Serverless = You do not manage the servers
 - Doesn't mean: there are no servers
- Stateless
- Triggers (Events, Schedule, Queue message, ...)
- Concurrency (with limit)
- Timeout (max 15 min.)
- Memory / CPU
 - Lambda allocates CPU in proportion to the amount of memory configured
- Layers (libraries - dependencies)



AWS Lambda

Identity and Access management (IAM)

IAM

- Users
- Groups
- Policies = Permissions
- Roles
 - Similar to an IAM user but provides temporary credentials that can be assumed by users, applications, or services to access AWS resources
- EC2 instance profile
 - IAM Role attached to EC2 instance





IAM Policy Example

- Allow these actions on all EC2 resources
- Any IAM user/group/role with this policy attached has the permissions to do these actions on all EC2 instances

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "ec2:DescribeInstances",
      "ec2:DescribeImages",
      "ec2:DescribeTags",
      "ec2:DescribeSnapshots"
    ],
    "Resource": "*"
  }]
}
```

Simple Storage Service (S3)



S3

A Highly scalable object storage service for storing and retrieving any amount of data

- **Bucket:** a container for objects (files / images / videos / ...)
- **Objects:** the actual files and their metadata (content-type / size / ...)
 - Key (path): <https://<bucket-name>.s3.<region>.amazonaws.com/images/img1.jpg>



S3 Storage Classes

S3 Standard

- general purpose frequently accessed data

S3 Standard IA (Infrequent Access)

- data you don't access often, but when you need it, you get it immediately
- can be used for disaster recovery, backups, ...
- min storage duration: 30 days



S3 Storage Classes (cont.)

S3 One Zone IA

- stored in 1 AZ = lower cost
- can be used to store secondary backups or data you can recreate
- min storage duration: 30 days

S3 Intelligent Tiering

- optimize cost for data with unknown or changing access patterns
- min storage duration: 30 days



S3 Storage Classes (cont.)

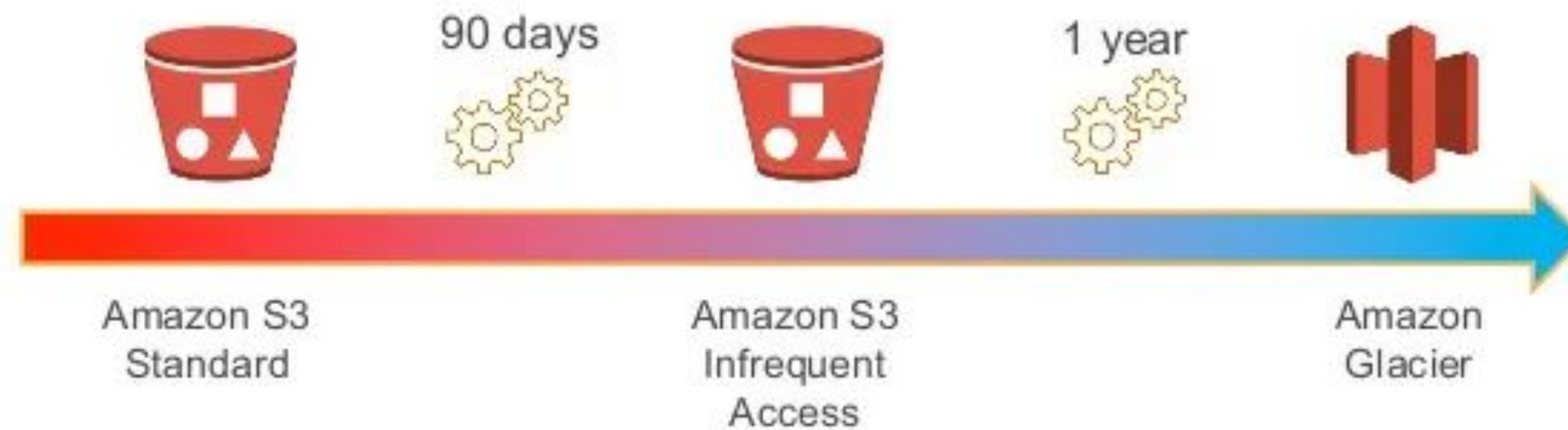
S3 Glacier

- cheap, but you don't access data immediately (mins or hours)
- min storage duration: 90 days

S3 Glacier Deep Archive

- cheapest (hours)
- min storage duration: 180 days

S3 Lifecycle Policy



<https://medium.com/@zixinhuang/s3-lifecycle-management-in-aws-8889085be66b>



S3 Features

Versioning

- keep multiple versions of an object in one bucket so that you can restore objects that are accidentally deleted or overwritten

Replication

- enable automatic, asynchronous copying of objects across S3 buckets
- buckets can be in different regions and owned by different AWS accounts



S3 Features (cont.)

Encryption

- Server-side encryption: Amazon S3 encrypts your objects before saving them on disks in AWS data centers and then decrypts the objects when you download them
- Client-side encryption: You encrypt your data client-side and upload the encrypted data to Amazon S3. In this case, you manage the encryption process, encryption keys, and related tools.

Databases

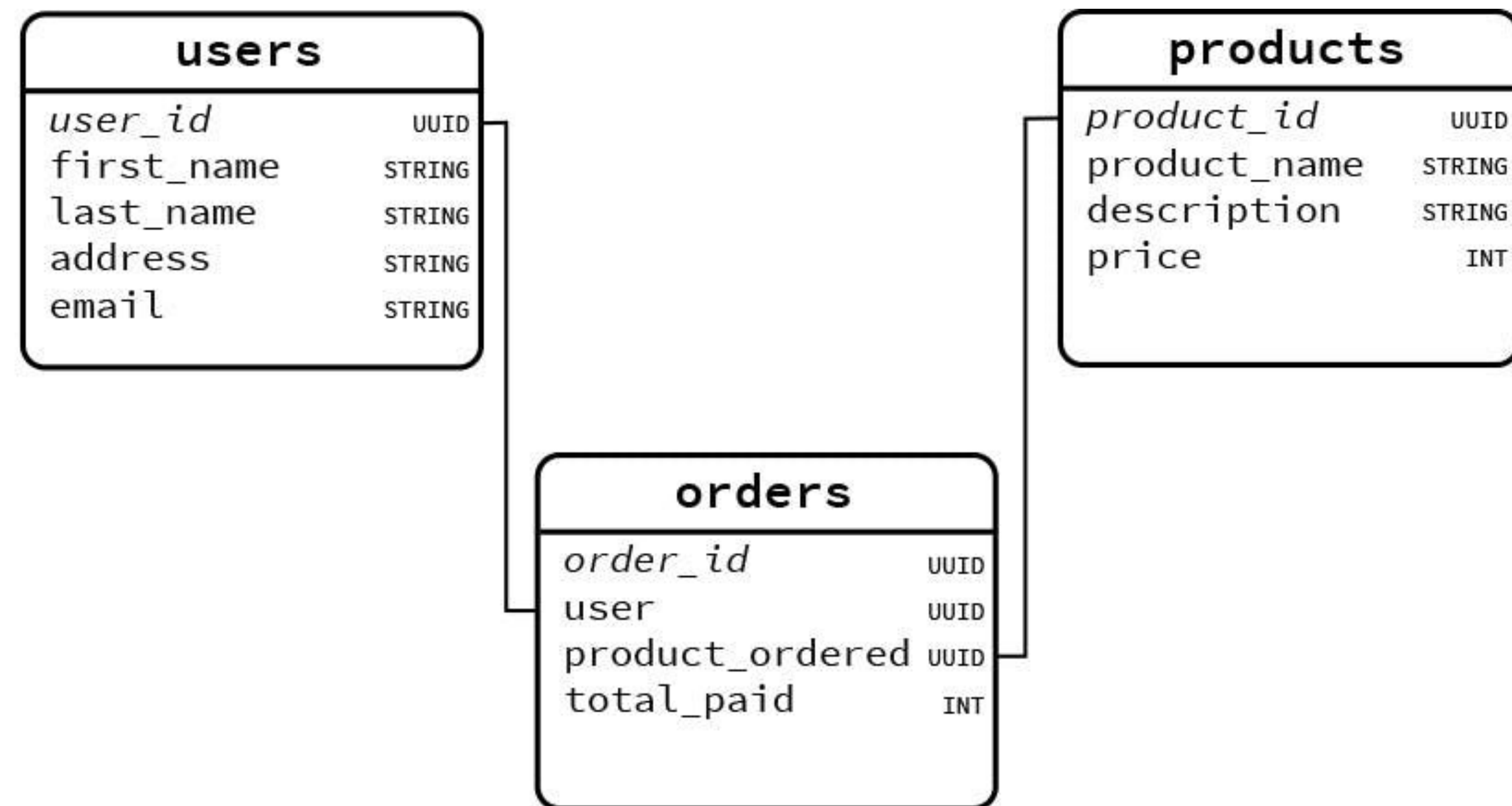


Database Types

- SQL / Relational Databases = **AWS RDS**
- NoSQL Databases = **AWS DynamoDB**

RDS

- Managed Relational DB
- DB Engines:
 - Microsoft SQL Server
 - MySQL
 - PostgreSQL
 - MariaDB
 - Oracle Database
 - IBM Db2



cockroachlabs.com



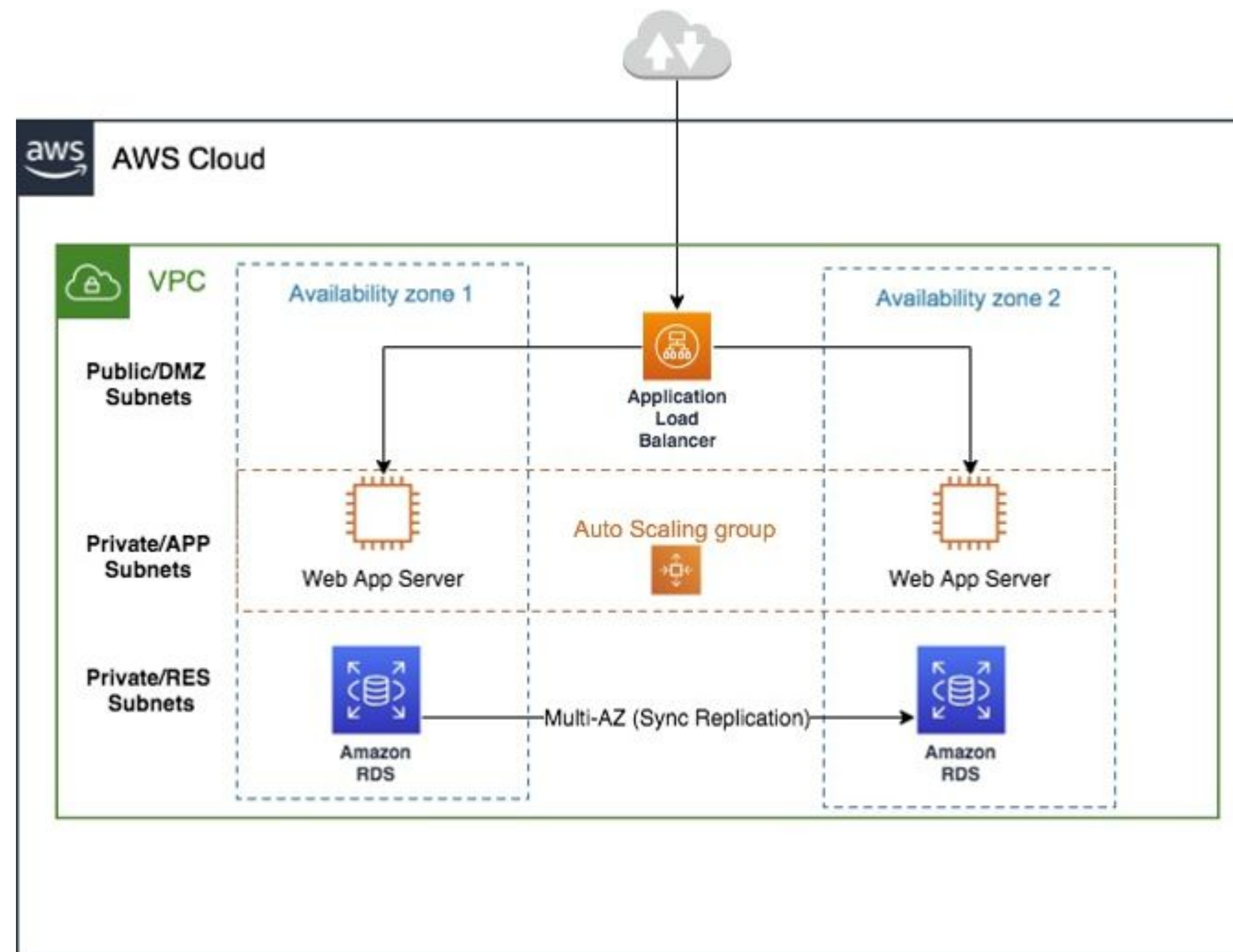
DynamoDB

- Managed NoSQL DB
 - Key / Value
 - JSON documents
- Tables / Partitions (Scaling)
- DynamoDB Accelerator (DAX) = in-memory caching
- Replication (multiple AZs / regions)

```
{
  "books": [
    {
      "id": "01",
      "language": "Java",
      "author": "H. Javeson",
      "year": 2015
    },
    {
      "id": "07",
      "language": "C++",
      "edition": "second",
      "author": "E. Sepp",
      "price": 22.25
    }
  ]
}
```

cs186berkeley.net

Three-tier Web Application on AWS



CloudFormation



CloudFormation

- Infrastructure as Code (IaC)
 - “Write once, deploy many times.”
 - Automated deployments
= no AWS console clicks
 - Version-controlled
 - Rollback on failure

```
AWSTemplateFormatVersion: "2010-09-09"
Description: Simple EC2 Example - Hardcoded Configuration

Resources:

  MySecurityGroup:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: Allow SSH access
      SecurityGroupIngress:
        - IpProtocol: tcp
          FromPort: 22
          ToPort: 22
          CidrIp: 0.0.0.0/0

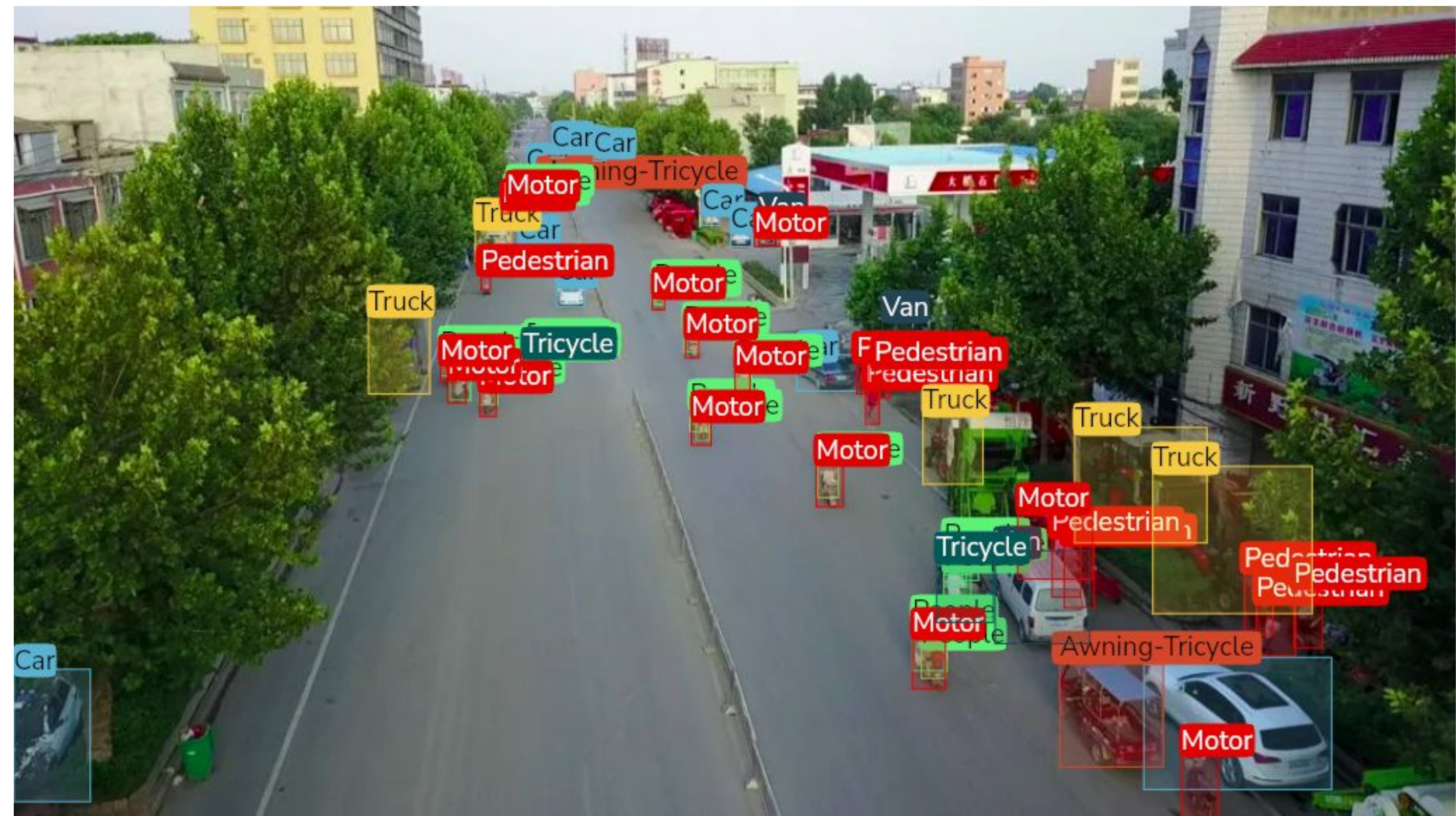
  MyEC2Instance:
    Type: AWS::EC2::Instance
    Properties:
      InstanceType: t2.micro
      KeyName: my-keypair-name # Replace with your key pair
      ImageId: ami-0c02fb55956c7d316 # Amazon Linux 2 (us-east-1)
      SecurityGroups:
        - !Ref MySecurityGroup
      Tags:
        - Key: Name
          Value: CFN-Demo-Instance
```



AI & ML Services

Amazon Rekognition

- Image and Video analysis
 - Object Detection
 - Facial Analysis
 - Unsafe Content Detection
 - Text Detection



<https://www.picsellia.com/post/object-detection-datasets>



Amazon Rekognition

Resource Groups ▼ ★

🔔 Brien Posey ▼ N. Virginia ▼ Support ▼

Facial analysis

Get a complete analysis of facial attributes, including confidence scores.



Done with the demo?
[Learn more](#)

▼ Results



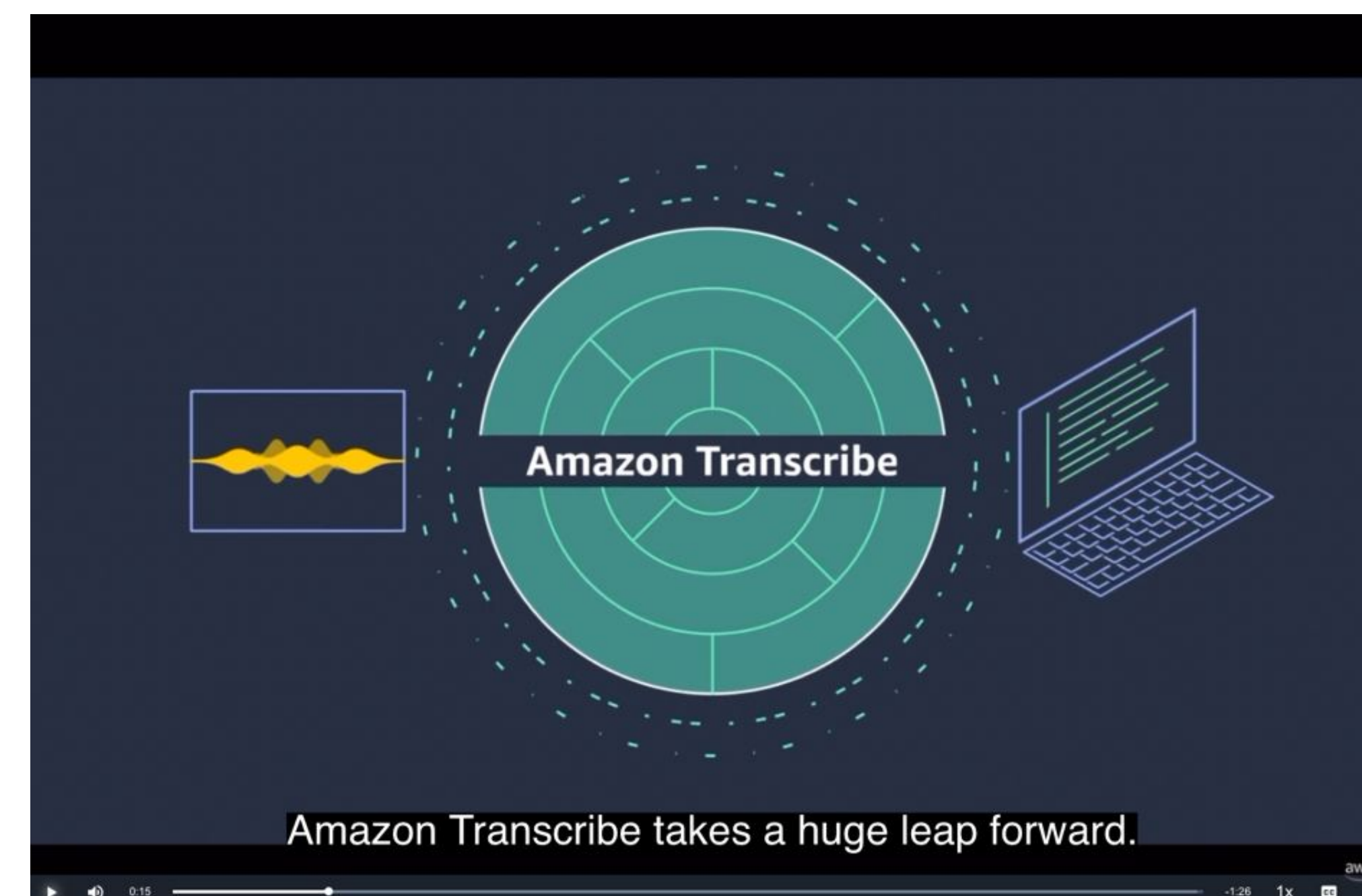
looks like a face	99.9 %
appears to be male	99.9 %
age range	35 - 52 years old
smiling	96.2 %
appears to be happy	99.8 %
wearing glasses	97.7 %

<https://virtualizationreview.com/articles/2018/02/20/amazon-rekognition-api-for-image-analysis.aspx>

Amazon Transcribe

- Speech to Text

- Real time (streaming) or media files (mp3, mp4, ...)
- Supports 50+ languages
- Custom Language Model
- Filter sensitive information (PII)
- Use Cases:
 - Video Subtitles
 - Customer support



Amazon Polly

- Text to Speech

- Real time (streaming) or documents (pdf, web page ...)
- Offers 100+ male and female voices in 40+ languages

Text-to-Speech

Listen, customize, and download speech. Integrate when you're ready.

Type or paste your text in the window, choose your language and region, choose a voice, choose Listen to speech, and then integrate it into your applications and services.

Plain text

SSML

?

Hi! My name is Jeff. I will read any text you type here.

944 characters remaining (1000 maximum)

Show default text

Clear text

Language and Region

English, US

Voice

☐ Joanna, Female

☐ Salli, Female

☐ Kimberly, Female

☐ Kendra, Female

☐ Ivy, Female

☐ Justin, Male

☒ Joey, Male

Listen to speech

Save speech as MP3

Change file format

Amazon Textract

- Automatically extract text, handwriting, and data from any document

Amazon Textract > Analyze document

Analyze document [info](#)

Download results

Upload document

Drag or upload a document to see its text, form data (key-value pairs and selection elements), and table data.

Sample document

Employment Application

Application Information

Full Name: Jane Doe

Phone Number: 555-0100

Home Address: 23 Any Street, Any Town, USA

Mailing Address: same as above

How did you hear about this position?

☒ Job fair

☐ Website

☐ Company Employee

Previous Employment History

Start Date	End Date	Employer Name	Position Held	Reason for leaving
1/15/2009	6/30/2011	Any Company	Assistant baker	relocated
7/1/2011	8/10/2013	Example Corp.	Baker	better opp.
8/10/2013	Present	Any Company	head baker	N/A, current

Raw textFormsTablesHuman review

Search

Full Name:

Jane Doe

Phone Number:

555-0100

Home Address:

123 Any Street, Any Town, USA

Mailing Address:

same as above

Job fair

SELECTED

Website

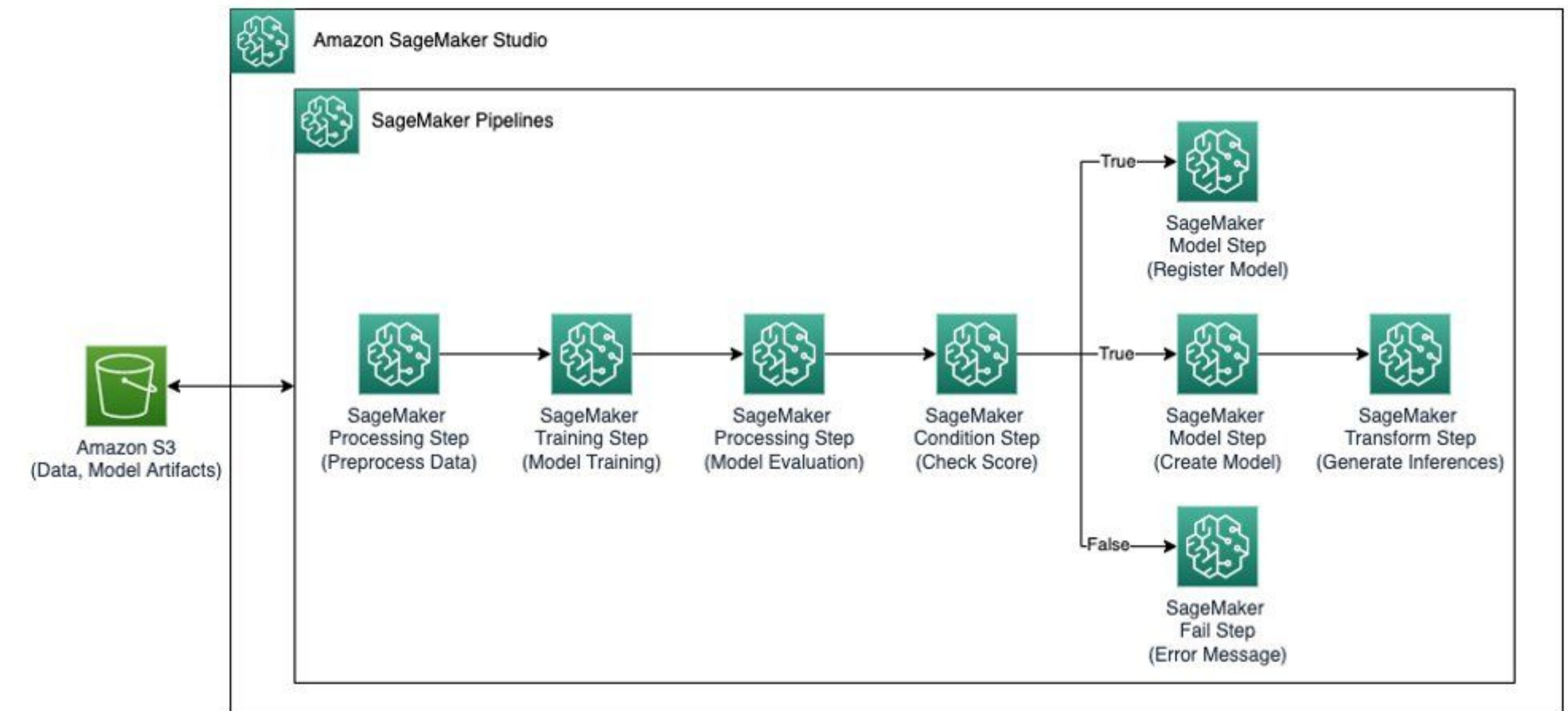
NOT_SELECTED

Company Employee

NOT_SELECTED

Amazon SageMaker

- Build, Train and Deploy ML models
 - Data preparation and processing
 - Model training and optimization
 - Real-time and batch inference
 - Build training/inference pipelines
 - Integrated development environment (supports Jupyter Notebooks)





Amazon Bedrock

- **Build GenAI applications** using a wide range of foundation models
 - Amazon Nova
 - Amazon Titan
 - Anthropic Claude
 - DeepSeek-R1
 - Cohere Command & Embed
 - Meta Llama
 - others...

Q&A

Thank You!

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